Modified title: Equation Playground and Log Page

December 9, 2025

Contents

Figure 1: System overview of the proposed method's pipeline

Abstract

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa. |Hello

1 Basic Algebra

A simple quadratic equation:

$$ax^2 + bx + c = 0, \tag{1}$$

with solutions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \tag{2}$$

Binomial theorem:

$$(x + y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k. \tag{3}$$

Geometric series:

$$\sum_{k=0}^{n} r^k = \frac{1 - r^{n+1}}{1 - r}, \qquad r \neq 1. \tag{4}$$

$$(a+b)^2 = a^2 + 2ab + b^2, \tag{5}$$
$$(a-b)^2 = a^2 - 2ab + b^2, \tag{6}$$
$$a^2 - b^2 = (a-b)(a+b). \tag{7}$$

2 Calculus

Definition of the derivative:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}. \tag{8}$$

Fundamental theorem of calculus:

$$\int_a^b f'(x)\,dx = f(b) - f(a). \tag{9}$$

Some common derivatives:

$$\frac{d}{dx} x^n = n x^{n-1}, \tag{10}$$
$$\frac{d}{dx} e^x = e^x, \tag{11}$$
$$\frac{d}{dx} \sin x = \cos x, \tag{12}$$
$$\frac{d}{dx} \cos x = -\sin x. \tag{13}$$

Some common integrals:

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1, \tag{14}$$
$$\int \frac{1}{x}\,dx = \ln|x| + C, \tag{15}$$
$$\int e^x\,dx = e^x + C, \tag{16}$$
$$\int \sin x\,dx = -\cos x + C. \tag{17}$$

A multivariable integral:

$$\iint_D (x^2 + y^2)\,dA \tag{18}$$

over some domain $D \subset \mathbb{R}^2$.

3 Linear Algebra

Matrix equation:

$$A\mathbf{x} = \mathbf{b}, \tag{19}$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}. \tag{20}$$

Determinant of a 2×2 matrix:

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc. \tag{21}$$

Eigenvalue equation:

$$A\mathbf{v} = \lambda \mathbf{v}. \tag{22}$$

4 Probability and Statistics

Probability of event A:

$$P(A) = \frac{\text{number of favorable outcomes}}{\text{total number of outcomes}}. \tag{23}$$

Linearity of expectation:

$$\mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} \mathbb{E}[X_i]. \tag{24}$$

Variance:

$$\operatorname{Var}(X) = \mathbb{E}[X^2] - \big(\mathbb{E}[X]\big)^2. \tag{25}$$

Normal distribution pdf:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right). \tag{26}$$

Bayes' theorem:

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}. \tag{27}$$

5 Logarithms (Log Page)

Basic definition:

$$\log_a b = c \iff a^c = b, \tag{28}$$

for $a > 0$, $a \neq 1$, and $b > 0$.

Change of base:

$$\log_a b = \frac{\log_c b}{\log_c a}. \tag{29}$$

5.1 Logarithm Identities

For $a > 0$, $a \neq 1$, and $x, y > 0$:

$$\log_a(xy) =_a x + \log_a y, \tag{30}$$

$$\log_a \left(\frac{x}{y}\right) = \log_a x - \log_a y, \tag{31}$$

$$\log_a(x^k) = k \log_a x, \tag{32}$$

$$\log_a a = 1, \tag{33}$$

$$\log_a 1 = 0. \tag{34}$$

Natural logarithm:

$$\ln x = \log_e x. \tag{35}$$

Derivative of the natural logarithm:

$$\frac{d}{dx} \ln x = \frac{1}{x}. \tag{36}$$

Derivative of a general logarithm:

$$\frac{d}{dx} \log_a x = \frac{1}{x \ln a}. \tag{37}$$

An integral involving logarithms:

$$\int \ln x \, dx = x \ln x - x + C. \tag{38}$$

5.2 Logarithmic Equations

Solve

$$\log_2(x - 1) + \log_2(x + 3) = 4. \tag{39}$$

Using the product rule:

$$\log_2 \big((x - 1)(x + 3)\big) = 4 \implies (x - 1)(x + 3) = 2^4 = 16. \tag{40}$$

Thus

$$x^2 + 2x - 3 = 16 \implies x^2 + 2x - 19 = 0. \tag{41}$$

5.3 Logarithmic Series

For $|x| < 1$:

$$\ln(1 + x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}. \tag{42}$$

For $x > -1$, $x \neq 0$:

$$\ln x = 2 \sum_{n=0}^{\infty} \frac{1}{2n + 1} \left(\frac{x - 1}{x + 1} \right)^{2n+1}. \tag{43}$$

5.4 A Long Log Page Filler

Here is some filler text to make this a long page:

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetuer id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.